

Executive Business Report: Sales Forecasting and Demand Analytics

1. Executive Summary

The analysis shows that sales have generally strengthened over the four-year period, although demand varies substantially by month, product group and region. Among the three forecasting approaches tested XGBoost produced the lowest forecasting errors and is therefore the preferred model for operational planning. The next three monthly sales totals are forecast at approximately \$47.8K, \$69.8K and \$44.1K. The analysis also identified unusual high and low-sales weeks and separated product sub-categories into four demand groups, allowing inventory policies to be matched more closely to actual demand behaviour. The main business opportunity is to use the forecast and demand segments together to improve replenishment, reduce avoidable stockouts and limit excess inventory.

2. Key Findings from Sales Analysis and Forecasting

- Annual sales increased strongly in the later years of the dataset, with 2018 recording the highest annual sales. Monthly sales show recurring fluctuations as well as an overall upward long-term trend.
- The East region showed the most consistent annual sales growth, with the lowest variation in year-to-year growth (standard deviation about 1.79 percentage points).
- XGBoost was the strongest forecasting model: MAE \$10,604.92, RMSE \$13,309.49 and MAPE 14.59%. It outperformed SARIMA (MAPE 19.38%) and Prophet (MAPE 23.87%) on the common holdout period.
- For the segment forecasts, the East region showed the strongest upcoming percentage growth, rising from roughly \$12K in the first forecast month to about \$15K in the third month (approximately 25%).

3. Three-Month Sales Outlook

The preferred XGBoost model forecasts total sales of \$47,826 in Month 1, \$69,796 in Month 2 and \$44,112 in Month 3. Month 2 is expected to be the strongest of the three, followed by a moderation in Month 3.

Because XGBoost does not produce a built-in statistical confidence interval in this implementation, the ranges below are indicative uncertainty bands based on the model's historical RMSE rather than formal confidence intervals:

Period	Forecast	Indicative range
Month 1	\$47,826	\$21,739 to \$73,912
Month 2	\$69,796	\$43,709 to \$95,882
Month 3	\$44,112	\$18,026 to \$70,199

4. Top Anomalies and Possible Business Causes

- **22 Mar 2015 (\$37,704):** The largest detected spike. A plausible cause is a large customer order, a short promotional campaign or a concentrated batch of high-value purchases.
- **22 Nov 2015 (\$26,794):** The timing is consistent with a possible pre-holiday or promotional sales uplift. Promotion data would be required to confirm the cause.
- **9 Aug 2015 (\$25,207):** This unusually high week may reflect a campaign, bulk order or temporary demand surge. The sales dataset alone cannot establish the exact cause.

5. Product Demand Segments and Stocking Strategy

- **High Volume, Stable Demand:** Maintain relatively high stock availability and regular replenishment to reduce stockout risk.
- **Low Volume, Stable Demand:** Use leaner inventory and smaller replenishment quantities to reduce carrying costs.
- **Rapidly Growing Demand:** Increase stock progressively and review demand frequently. Growth should be monitored to ensure that a high percentage increase is not simply caused by a low starting base.
- **High-Value, Volatile Demand:** Use flexible replenishment, careful safety-stock limits and frequent review because excess inventory is costly while demand is less predictable.

6. Three Business Recommendations

- Use XGBoost as the primary planning forecast. It achieved the lowest error across all three evaluation measures: MAE \$10.6K, RMSE \$13.3K and MAPE 14.59%. Use its monthly forecast as the baseline for purchasing and cash-flow planning, with human review for major promotions or known events.
- Prioritize inventory capacity for growing demand, especially the East region. The East region showed the strongest upcoming segment growth at roughly 25% from the first to the third forecast month. Supply planning should review reorder levels and supplier lead times for products driving this growth.
- Replace one-size-fits-all stocking with cluster-based inventory rules. The four demand clusters show materially different combinations of sales volume, growth and volatility. Stable high-volume products should receive stronger availability targets, while volatile high-value products should be managed with tighter review cycles and controlled safety stock.

7. Risk and Limitation

The system is trained on only four years of historical sales and does not include important external drivers such as promotions, prices, stockouts, holidays, competitor actions or economic conditions. As a result, the model may treat one-off events as repeatable patterns and cannot explain the true cause of an anomaly. Forecasts should therefore support—not replace—commercial and supply-chain judgement, and model performance should be monitored as new data becomes available.